

JAVA LAB RECORD

Experiment No.	06
Date	23/09/26
Title	Implementation of LinkedHashSet Methods in Java

Objective / Aim

To study and implement commonly used methods of the Java LinkedHashSet class, including adding unique elements, checking for elements, finding the size, removing elements, and clearing all elements while maintaining insertion order.

System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Keyboard and mouse
- Monitor/display

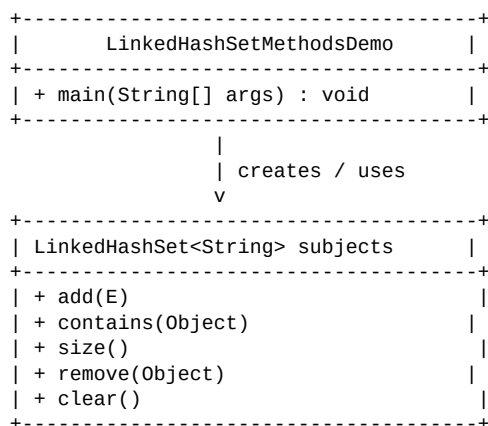
Software Requirements

- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK): 17 or later
- IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Start the program.
2. Import the `LinkedHashSet` class from `java.util`.
3. Create a `LinkedHashSet` named `subjects` to store `String` values.
4. Add `Java`, `Python`, and `C++` using `add(E e)`.
5. Attempt to add `Java` again and observe that duplicate elements are not stored.
6. Check whether `Java` and `HTML` are present using `contains(Object)`.
7. Find the number of elements using `size()`.
8. Remove `Python` using `remove(Object)`.
9. Clear all elements using `clear()`.
10. Find the size of the `LinkedHashSet` after clearing it.
11. Display all results.
12. Stop the program.

UML Class Diagram



Source Code

```

import java.util.LinkedHashSet;

class LinkedHashSetMethodsDemo {

    public static void main(String[] args) {

        // Creating a LinkedHashSet
        LinkedHashSet<String> subjects = new LinkedHashSet<>();

        // add(E e)
        subjects.add("Java");
        subjects.add("Python");
        subjects.add("C++");
        subjects.add("Java");    // Duplicate, will not be added

        System.out.println("After add(): " + subjects);

        // contains(Object o)
        System.out.println("Contains Java? " + subjects.contains("Java"));
        System.out.println("Contains HTML? " + subjects.contains("HTML"));

        // size()
        System.out.println("Size of LinkedHashSet: " + subjects.size());

        // remove(Object o)
        subjects.remove("Python");
        System.out.println("After remove(): " + subjects);

        // clear()
        subjects.clear();
        System.out.println("After clear(): " + subjects);

        // Checking size after clear
        System.out.println("Size after clear(): " + subjects.size());
    }
}

```

Completion and Execution Commands

1. Save the program as:
LinkedHashSetMethodsDemo.java
2. Open Command Prompt / Terminal and move to the folder containing the Java file.
3. Compile the program:
javac LinkedHashSetMethodsDemo.java
4. Execute the program:
java LinkedHashSetMethodsDemo
5. The results of all LinkedHashSet operations are displayed on the console.

Output

```

After add(): [Java, Python, C++]
Contains Java? true
Contains HTML? false
Size of LinkedHashSet: 3
After remove(): [Java, C++]
After clear(): []
Size after clear(): 0

```

Result

Thus, the Java program was successfully implemented and executed to demonstrate the commonly used methods of the LinkedHashSet class, including unique element storage and preservation of insertion order.